

Odd cycles, sech spectra, and a square-tail inequality

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Abstract

Let π be a uniform permutation of $\{1, \dots, 2r-1\}$, let $o(\pi)$ be its number of odd cycles, and put $K_r = (o(\pi) - 1)/2$ and $\Pi_r(u) = \mathbb{E}[u^{K_r}]$. The polynomial Π_r has three exact descriptions: the normalized Gram determinant $\det(uI + G_r)/(2r-1)!$, a product of Bernoulli factors read off the positive spectrum of the consecutive-integer Jacobi matrix, and the cardinality generating function of a determinantal point process with an explicit integer configuration law. Between them they account for the one-row facts used here: real-rootedness, coefficient log-concavity and the Poisson–binomial law.

This paper proves a quantitative comparison of adjacent rows. Writing $L(f)_t = f_t^2 - f_{t-1}f_{t+1}$ for the coefficientwise Turán operator, $S_{r,t} = L((1+u)\Pi_r)_t$, $V_{r,t} = L(\Pi_r)_{t-1}$ and $d_{r,t} = S_{r,t} - \left(\frac{2r-1}{2r}\right)^2 S_{r-1,t}$, the main theorem is the *square-tail inequality*

$$d_{r,t}^2 \geq \frac{2}{r^2} S_{r,t} V_{r,t}, \quad r \geq 2, \quad 2 \leq t \leq r.$$

The same machinery also orients the decrement, $d_{r,t} > 0$; the content of the square-tail theorem is the factor 2, which Newton’s inequalities do not pay. That factor is the structural constant, not the extremal one. A second theorem identifies the optimal uniform constant as $C_\star = 2\rho(129, 2) = 2.029152329129741\dots$, an explicit rational whose reduced numerator and denominator have 1,233 digits each, attained at the single cell $(129, 2)$ and at no other. The proof establishes monotonicity and discrete convexity for factorially scaled coefficient ratios, proves a uniform decrement bound, and shows that a linear lowering profile is extremal. Seven complete fixed columns and a 500-row exact core leave two unbounded symbolic regions. The two upper cells collapse to elementary coefficient identities, while the adjacent lower stripe $t = 1$ contains exact counterexamples. Thus $2 \leq t \leq r$ is the exact uniform nontrivial range, not a proof margin. As an application the inequality settles a boundary problem for the shift-wall rows. The sech/odd-cycle/Meixner bridge underlying the setup is classical, and §3 says exactly where it comes from; what is new is the coefficient-index layer that the inequalities live on.

1 Odd cycles and the cross-row question

Odd-cycle polynomials sit at once in permutation enumeration, orthogonal-polynomial theory and determinantal probability. For a uniform $\pi \in S_{2r-1}$ the number of odd cycles $o(\pi)$ has the parity of $2r - 1$ and is therefore odd, so

$$K_r = \frac{o(\pi) - 1}{2}$$

is an integer in $\{0, \dots, r-1\}$, and we set $\Pi_r(u) = \mathbb{E}[u^{K_r}]$. We use $\Pi_1(u) = 1$, the row of the unique permutation in S_1 , when a predecessor is needed at $r = 2$. The first two nontrivial rows are

$$\Pi_2(u) = \frac{5+u}{6}, \quad \Pi_3(u) = \frac{89+30u+u^2}{120}.$$

These are not isolated combinatorial data. Section 2 proves that $F_r = (2r-1)! \Pi_r$ satisfies

$$F_r(u) = \det(uI + G_r) = \prod_{j=1}^{r-1} (u + \xi_{r,j}^2),$$

where G_r is an integer Gram matrix and the $\xi_{r,j}$ are the positive eigenvalues of the Jacobi matrix with off-diagonal entries $1, 2, \dots, 2r-2$; equivalently, K_r has the cardinality law of a determinantal point process with kernel $(I + G_r)^{-1}$, and §2.4 realizes that law by an explicit map. Section 3 traces the classical sech/Jacobi and odd-cycle components in the literature.

Each of those descriptions concerns one row at a time. The question here is what happens to coefficient curvature when the row is lowered by one. For a polynomial $f(u) = \sum_k f_k u^k$ write

$$L(f)_t = f_t^2 - f_{t-1} f_{t+1}, \tag{1}$$

with coefficients outside the row understood to be zero, and put

$$S_{r,t} = L((1+u)\Pi_r)_t, \quad V_{r,t} = L(\Pi_r)_{t-1}, \quad d_{r,t} = S_{r,t} - \left(\frac{2r-1}{2r}\right)^2 S_{r-1,t}. \tag{2}$$

Strict degree-aware Newton inequalities give $S_{r,t}, V_{r,t} > 0$ on the range below. The adjacent-row decrement is positive on the interior, while the two upper cells are explicit. Neither fact determines the *size* of the decrement. The main theorem does.

Theorem 1.1 (square-tail inequality). *For every pair of integers $r \geq 2$ and $2 \leq t \leq r$,*

$$d_{r,t}^2 \geq \frac{2}{r^2} S_{r,t} V_{r,t}.$$

Equivalently, with

$$\rho(r, t) = \frac{r^2 d_{r,t}^2}{2 S_{r,t} V_{r,t}}, \tag{3}$$

Theorem 1.1 is exactly $\rho \geq 1$ on that domain.

The factor 2 is clean and structural, and it is what the proof below pays for directly. It is not extremal. The extremal constant is identified separately.

Theorem 1.2 (optimal uniform constant). *Put $C_\star = 2\rho(129, 2)$. For every pair of integers $r \geq 2$ and $2 \leq t \leq r$,*

$$d_{r,t}^2 \geq \frac{C_\star}{r^2} S_{r,t} V_{r,t},$$

with equality if and only if $(r, t) = (129, 2)$. Equivalently $\min \rho = \rho(129, 2)$, attained at exactly one cell, and no constant larger than $C_\star$ is admissible.

Theorem 1.2 implies Theorem 1.1, since $C_\star > 2$. The two use separately gated certificate chains, but share definitions, analytic machinery, and the finite core; they are not claimed as fully independent implementations. Theorem 1.1 is retained because its proof is the structural one, and because the slack it leaves is uniform and explicit:

$$\rho(r, t) - 1 \geq \rho(129, 2) - 1 = 0.014576164564870742 \dots,$$

with equality only at $(129, 2)$.

1.1 What the factor 2 costs

Three exact computations fix how much the classical structure can pay, and where the domain hypotheses come from.

- (a) *Newton is not enough.* Newton's inequalities applied to the two rows give $S_{r,t} \geq V_{r,t}$, strictly off $t = r$, which is the right direction and the wrong size. As a route to Theorem 1.1 it is refuted on the actual cell $(r, t) = (4, 2)$, where $S_{4,2} - 2V_{4,2} = -1931/26460$.
- (b) *Root data alone is not enough.* Interlacing, positivity of all power sums, the factorial normalization and $d_{r,t} > 0$ together do not imply the statement. Take the strictly alternating Bernoulli parameters

$$\theta_4 = \left(\frac{13}{30}, \frac{9}{70}, \frac{5}{1404} \right), \quad \theta_3 = \left(\frac{77}{180}, \frac{3}{154} \right),$$

ordered $\theta_{4,1} > \theta_{3,1} > \theta_{4,2} > \theta_{3,2} > \theta_{4,3}$, and set $\tilde{\Pi}_j(u) = \prod_i (1 - \theta_{j,i} + \theta_{j,i}u)$. The parameter products are $1/7!$ and $1/5!$, so the leading coefficients have the required normalization. Direct substitution gives $d_{4,2} = 526797418380619/8414884558080000 > 0$ and the exact ratio

$$\frac{277515520012484937028354823161}{315577547434843909307279254848} < 1.$$

Any proof must therefore use structure beyond those root data.

- (c) *The lower boundary is genuine.* The cell $(r, t) = (2, 1)$ fails, with exact deficit

$$d_{2,1}^2 - \frac{1}{2}S_{2,1}V_{2,1} = -\frac{4351}{20736}, \quad \text{from } S_{2,1} = \frac{31}{36}, \quad V_{2,1} = \frac{25}{36}, \quad d_{2,1} = \frac{43}{144},$$

and the corresponding $t = 1$ stripe in the shift-wall application fails on a contiguous window of rows exhibited in §7. Thus the lower bound $t \geq 2$ cannot be weakened in a uniform theorem. For $r \geq 3$ the two cells at the other end, $t = r - 1, r$, are both true by Proposition 6.4.

The exact cell $(r, t) = (129, 2)$ gives the witness

$$\rho(129, 2) = 1.014576164564870742 \dots,$$

so no constant larger than $2\rho(129, 2) = 2.029152 \dots$ can replace 2. This witness is the global minimizer, and it is the only one: see Theorem 1.2. The runner-up is the neighbouring cell $(128, 2)$, at $\rho(128, 2) = 1.014577752404662 \dots$, so the minimum is separated from the rest of the array by 1.5878×10^{-6} .

1.2 The proof in four moves

- (1) After the factorial scaling $b_{r,k} = (2k + 1)![u^k]F_r$, $X_{r,k} = b_{r,k}/b_{r,k-1}$, the ratios $X_{r,k}$ decrease in k and are discretely convex. Every decrement is therefore controlled by the first, and $0 \leq X_{r,k-1} - X_{r,k} < 5$ throughout (§5).
- (2) The preceding row is a coefficientwise lowering of the current one. Among all lowering profiles the curvature constraints allow, the linear comparator with the same first lowering maximizes the predecessor Turán determinant (§5.3).

- (3) Linear extremality reduces Theorem 1.1 to a correlated scalar inequality in four cell-local quantities. One estimate closes the top of every row; a second closes the joint window in which level and coefficient index grow together (§§5.3–6).
- (4) Seven fixed columns leave only $t \geq 9$. Outside the top region one then has $a_r > (t-2)^2 > 5t$, so the joint-window theorem applies; an exact full-row core through $r = 503$ completes the interior. The two upper cells are then closed by one wall-coefficient calculation (§6).

Moves (1)–(3) are exact analytic inequalities, coefficient signs and two Sturm checks, and they close both unbounded interior regions outright. Move (4) uses exact rational arithmetic only on the bounded residue: 125,250 full-row cells on $4 \leq r \leq 503$ and the remaining fixed-column prefixes. Appendix A maps the computational record.

2 Odd cycles, Jacobi matrices, and the determinantal point process

2.1 The parity split

For $n \geq 1$ let J_n be the *sech Jacobi matrix*: the real symmetric tridiagonal matrix on the basis $e_0, \dots, e_{n-1}$ with zero diagonal and

$$(J_n)_{k-1,k} = (J_n)_{k,k-1} = k, \quad 1 \leq k \leq n-1.$$

Fix $r \geq 2$ and put $n = 2r-1$. Ordering the even-indexed sites $e_0, e_2, \dots, e_{2r-2}$ before the odd-indexed sites $e_1, e_3, \dots, e_{2r-3}$ puts J_{2r-1} in the block form

$$J_{2r-1} \cong \begin{pmatrix} 0 & D_r \\ D_r^\top & 0 \end{pmatrix}, \quad (4)$$

where $D_r \in \mathbb{R}^{r \times (r-1)}$ is the bidiagonal incidence matrix of the parity split,

$$(D_r)_{a,b} = (2b-1) \mathbf{1}_{a=b} + 2b \mathbf{1}_{a=b+1}, \quad 1 \leq a \leq r, \quad 1 \leq b \leq r-1. \quad (5)$$

Write $G_r = D_r^\top D_r$, an $(r-1) \times (r-1)$ symmetric tridiagonal integer matrix with

$$(G_r)_{b,b} = (2b-1)^2 + (2b)^2, \quad (G_r)_{b,b+1} = 2b(2b+1). \quad (6)$$

Equations (5)–(6) use one-based matrix indices. In the DPP formulas below we relabel G_r by the sites $0, \dots, r-2$: site j corresponds to column $j+1$ of D_r .

Example 2.1 ($r = 2$ and $r = 3$). For $r = 2$,

$$D_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad G_2 = (5), \quad F_2(u) = \det(uI + G_2) = u + 5, \quad F_2(1) = 6 = 3!,$$

and $\Pi_2(u) = (5+u)/6$. Correspondingly $\text{spec}(J_3) = \{0, \pm\sqrt{5}\}$, so the single positive eigenvalue is $\xi_{2,1} = \sqrt{5}$ and $\theta_{2,1} = 1/(1+5) = 1/6$, matching the linear coefficient of Π_2 . For $r = 3$,

$$D_3 = \begin{pmatrix} 1 & 0 \\ 2 & 3 \\ 0 & 4 \end{pmatrix}, \quad G_3 = \begin{pmatrix} 5 & 6 \\ 6 & 25 \end{pmatrix}, \quad F_3(u) = u^2 + 30u + 89, \quad F_3(1) = 120 = 5!.$$

The three coefficients 89, 30, 1 count the permutations of $\{1, \dots, 5\}$ with 1, 3 and 5 odd cycles respectively.

2.2 Three exact representations

Let $M_n(z)$ be the *wall rows*

$$M_0 = 1, \quad M_1 = 1 + z, \quad M_n = M_{n-1} + n^2 z M_{n-2}, \quad \deg M_n = \lceil n/2 \rceil, \quad (7)$$

and put

$$F_r(u) = u^{r-1} M_{2r-2}(1/u) = (2r-1)! \Pi_r(u), \quad (8)$$

so that F_r is monic of degree $r-1$ with $F_r(1) = (2r-1)!$. Let $B_n = (I + J_n^2)^{-1}$.

We shall use the first two wall coefficients twice. The recurrence (7) gives them without a generating function.

Lemma 2.2 (terminal wall coefficients). *For every $n \geq 0$,*

$$[z]M_n = \frac{n(n+1)(2n+1)}{6}, \quad [z^2]M_n = \frac{n(n-2)(n-1)(n+1)(20n^2 - 8n - 21)}{360}.$$

Proof. Write $w_n = [z]M_n$ and $v_n = [z^2]M_n$. Since $[z^0]M_n = 1$, (7) gives

$$w_n = w_{n-1} + n^2, \quad v_n = v_{n-1} + n^2 w_{n-2}.$$

The displayed polynomials satisfy these recurrences and the initial values $w_0 = 0, w_1 = 1, v_0 = v_1 = 0$. $\square$

Theorem 2.3 (three faces of Π_r). *For every $r \geq 2$:*

- (i) combinatorial — $F_r(u) = \det(uI + G_r) = \sum_{\pi \in S_{2r-1}} u^{K_r(\pi)}$;
- (ii) spectral — *let $\pm \xi_{r,1}, \dots, \pm \xi_{r,r-1}$ together with 0 be the eigenvalues of J_{2r-1} , so that the $\xi_{r,j} > 0$ are the $r-1$ positive eigenvalues of J_{2r-1} , equivalently the positive zeros of its degree- $(2r-1)$ characteristic polynomial, and equivalently the positive square roots of the eigenvalues of G_r . Then, with $\theta_{r,j} = 1/(1 + \xi_{r,j}^2)$,*

$$\Pi_r(u) = \prod_{j=1}^{r-1} ((1 - \theta_{r,j}) + \theta_{r,j} u);$$

- (iii) determinantal — $\det(I + (u-1)B_{2r-1}) = u \Pi_r(u)^2$, while the smaller parity block

$$\mathcal{K}_r = (I + G_r)^{-1}$$

satisfies $\det(I + (u-1)\mathcal{K}_r) = \Pi_r(u)$.

Proof. The odd-cycle enumerator $A_n(y) = \sum_{\pi \in S_n} y^{o(\pi)}$ has exponential generating function $\exp(y \operatorname{artanh} x) / \sqrt{1-x^2}$; differentiating gives $(1-x^2)\partial_x A = (y+x)A$ and hence

$$A_0 = 1, \quad A_1 = y, \quad A_{n+1}(y) = y A_n(y) + n^2 A_{n-1}(y), \quad (9)$$

which is exactly the continuant recurrence of the consecutive-integer Jacobi matrix. Comparing (9) with (7) gives $A_n(y) = y^n M_{n-1}(y^{-2})$, so for $n = 2r-1$ we get $A_{2r-1}(y) = y F_r(y^2)$ and therefore $F_r(u) = \sum_{\pi} u^{K_r(\pi)}$, which is the second equality in (i). On the spectral side the characteristic polynomial $p_n(X) = \det(XI - J_n)$ obeys $p_{n+1} = X p_n - n^2 p_{n-1}$, so $p_n(X) = X^n M_{n-1}(-1/X^2)$; in particular p_{2r-1} is odd, its zero set is $\{0, \pm \xi_{r,1}, \dots, \pm \xi_{r,r-1}\}$, and by (4) the matrix J_{2r-1}^2 has parity blocks $D_r D_r^\top$ and $D_r^\top D_r = G_r$, whose common

nonzero spectrum is $\{\xi_{r,j}^2\}$. Hence $F_r(u) = \prod_j (u + \xi_{r,j}^2) = \det(uI + G_r)$, which is the first equality in (i). Dividing by $F_r(1) = (2r-1)!$ and writing $\theta_{r,j} = 1/(1 + \xi_{r,j}^2)$ gives (ii).

For (iii), $\det(I + (u-1)K_r) = \det(uI + G_r) / \det(I + G_r) = \Pi_r(u)$ directly. The full resolvent splits orthogonally as $B_{2r-1} \cong (I + D_r D_r^\top)^{-1} \oplus (I + D_r^\top D_r)^{-1}$; the first block has spectrum $\{1, \theta_{r,1}, \dots, \theta_{r,r-1}\}$ and the second $\{\theta_{r,1}, \dots, \theta_{r,r-1}\}$, so $\det(I + (u-1)B_{2r-1}) = u \Pi_r(u)^2$. $\square$

Everything in Theorem 2.3 follows from (9) and the parity split, and holds for every r , not on a checked range.

2.3 The configuration law

Part (iii) says K_r is equidistributed with the cardinality of the determinantal point process on $r-1$ sites with correlation kernel K_r . That process has an exact integer configuration law, which is sharper than the cardinality statement and is what makes the determinantal face checkable by hand.

Corollary 2.4. *Let X be the determinantal point process on $\{0, \dots, r-2\}$ with correlation kernel $K_r = (I + G_r)^{-1}$, equivalently with L -ensemble kernel G_r^{-1} . Then for every $S \subseteq \{0, \dots, r-2\}$,*

$$\Pr(X = S) = \frac{\det((G_r)_{S^c, S^c})}{(2r-1)!}, \quad (10)$$

with $\det(\emptyset) = 1$, and consequently

$$\#\{\pi \in S_{2r-1} : K_r(\pi) = k\} = \sum_{|S|=k} \det((G_r)_{S^c, S^c}). \quad (11)$$

Proof. The L -ensemble with kernel $L = G_r^{-1}$ has correlation kernel $L(I+L)^{-1} = (I+G_r)^{-1} = K_r$ and $\Pr(X = S) = \det(L_{S,S}) / \det(I+L)$. Jacobi's complementary-minor identity gives $\det(L_{S,S}) = \det((G_r^{-1})_{S,S}) = \det((G_r)_{S^c, S^c}) / \det G_r$, and $\det(I+L) = \det(I+G_r) / \det G_r$, so the two determinants of G_r cancel and $\det(I+G_r) = F_r(1) = (2r-1)!$ supplies the stated normalization. Summing (10) over $|S| = k$ and using Theorem 2.3(i) gives (11). $\square$

Example 2.5 ($r = 3$, by hand). With G_3 as in Example 2.1 and sites $\{0, 1\}$, the four complementary minors are

$$\det(G_3) = 89, \quad (G_3)_{1,1} = 25, \quad (G_3)_{0,0} = 5, \quad \det(\emptyset) = 1,$$

attached to $S = \emptyset, \{0\}, \{1\}, \{0, 1\}$ respectively. They sum to $120 = 5!$, and aggregating by $|S|$ returns 89, 30, 1, which are the coefficients of F_3 as they must be by (11).

2.4 The coupling

Corollary 2.4 is an equidistribution statement. It is realized by an explicit map, which we now give: the determinantal configuration is a *matching statistic* of the permutation.

Let Path_n be the path on vertices $1, \dots, n$ and give edge $\{j, j+1\}$ a set of j^2 colours. By a *coloured matching* of Path_n we mean a matching together with a colour for each of its edges; the unmatched vertices are its *monomers*.

Theorem 2.6 (peeling bijection). *There is a bijection between S_n and the coloured matchings of Path_n under which the number of monomers equals $o(\pi)$.*

Proof. Peel the maximum. If $\pi(n) = n$, delete n and leave path vertex n a monomer. Otherwise put $a = \pi^{-1}(n)$ and $b = \pi(n)$; delete the two letters a and n , bypassing them in their cycle so that $\pi^{-1}(a) \mapsto b$, standardize the result to a permutation of $[n-2]$, and place a dimer on the terminal edge $\{n-1, n\}$ coloured by the pair $(a, b) \in [n-1]^2$. This is the combinatorial reading of

$$A_n(y) = y A_{n-1}(y) + (n-1)^2 A_{n-2}(y),$$

and it is reversible as follows. Given the smaller permutation σ , let $E = [n] \setminus \{a, n\}$, let $\phi : E \rightarrow [n-2]$ be the increasing bijection, and unstandardize to $\tau = \phi^{-1}\sigma\phi$ on E . If $b = a$, adjoin the transposition $a \leftrightarrow n$. If $b \neq a$, then $b \in E$; with $c = \tau^{-1}(b)$, replace the arrow $c \mapsto b$ by $c \mapsto a \mapsto n \mapsto b$. These two cases uniquely recover the original permutation and are visibly inverse to the deletion. Deleting a fixed point removes one odd cycle; deleting the two-letter segment (a, n) from its cycle drops the length by 2 and so preserves its parity. Hence each monomer accounts for exactly one odd cycle. $\square$

Now take $n = 2r - 1$, so Path_n has r odd and $r - 1$ even vertices, and set

$$S(\pi) = \{b \in \{0, \dots, r-2\} : 2b+2 \text{ is an unmatched even vertex}\}. \quad (12)$$

Theorem 2.7 (the configuration is a matching statistic). *For every $r \geq 2$ and every $S \subseteq \{0, \dots, r-2\}$,*

$$|S(\pi)| = K_r(\pi) \quad \text{and} \quad \#\{\pi \in S_{2r-1} : S(\pi) = S\} = \det((G_r)_{S^c, S^c}).$$

Consequently $\Pr(S(\pi) = S)$ is the law of Corollary 2.4, and (12) couples the uniform permutation to the determinantal configuration on one probability space.

Proof. Every edge of Path_n joins an odd vertex to an even one, so each dimer covers one of each. Writing $o = o(\pi)$ for the number of monomers, the monomers at positions $p_1 < \dots < p_o$ must leave gaps of even length, so $p_1 - 1$, $p_{i+1} - p_i - 1$ and $n - p_o$ are all even; hence p_1 is odd, the parities alternate, and p_o is odd. Therefore o is odd and exactly $(o-1)/2$ of the monomers sit at even vertices, which is the first claim.

For the second, fix S and let $T = S^c$, so the matched even vertices are exactly those indexed by T . For each odd row set R with $|R| = |T|$, the bipartite graph underlying $D_{R,T}$ is a disjoint union of paths. It has at most one perfect matching, since an endpoint forces its incident edge and induction continues down each component. Thus $\det(D_{R,T})$ is either zero or, up to sign, the product of the weights of the corresponding path matching. Its square is exactly the number of colour choices, because an edge $\{j, j+1\}$ has j^2 colours. Summing over R therefore counts precisely the coloured matchings with trace T , and Cauchy–Binet gives

$$\#\{\pi : S(\pi) = S\} = \sum_R \det(D_{R,T})^2 = \det((D_r^\top D_r)_{T,T}) = \det((G_r)_{T,T}). \quad \square$$

Remark 2.8. Because D_r is bidiagonal, the minors $\det(D_{R,T})$ appearing above are indexed by a single switch point on each maximal run of T , which makes Theorem 2.7 checkable by hand on small rows. The weighted-matching reading of these continuants is classical Cayley–continuant territory [1]; the step taken here is the projection onto unmatched *even* vertices, which is what turns it into the determinantal coupling.

Remark 2.9. The inequalities of this paper still see only the cardinality of X and never its correlations; Theorem 2.7 is not used in the proof of Theorem 1.1.

2.5 Normalization, parity, and conjugation

Four nearby objects must be kept distinct.

Remark 2.10 (monic versus normalized). F_r is monic; Π_r is not. Π_r is normalized by $\Pi_r(1) = 1$ and has leading coefficient $1/(2r-1)!$. Monicity of F_r is structural, since $\det(uI + G_r)$ is monic by construction. Note also that $\det(uI + G_r)$ is not the characteristic polynomial χ_{G_r} of G_r but that polynomial evaluated at $-u$, up to sign: $\det(uI + G_r) = (-1)^{r-1} \chi_{G_r}(-u)$. Similarly, the object in Theorem 2.3(iii) is a cardinality-generating determinant $\det(I + (u-1) \cdot)$, not $\det(uI - \cdot)$.

Remark 2.11 (the parity blocks carry different laws). The odd-index block, of size $r-1$, has generating function Π_r and law K_r . The even-index block, of size r , has generating function $u\Pi_r$ and law $1 + K_r$. That extra factor u is a Bernoulli factor with parameter 1, that is, a point present with probability one. It exists because J_{2r-1} has odd size, which forces $\dim \ker J_{2r-1} = 1$; and it belongs to the even block rather than the odd one because the corresponding null vector is supported entirely on the even-index coordinates. The full $(2r-1)$ -point kernel has generating function $u\Pi_r^2$ and law $1 + K_r(\pi) + K_r(\sigma)$ for two *independent* uniform permutations. The blocks are therefore not interchangeable, and it is the smaller block K_r whose cardinality is K_r .

Remark 2.12 (equality in distribution). The Bernoulli decomposition in Theorem 2.3(ii) is an equality *in distribution*, on a separate probability space. It is not a pointwise decomposition of K_r into independent functions of one permutation.

Remark 2.13 (the deposited kernel is a conjugate). The kernel released in panel two [4] is not B_n but its signature conjugate

$$\tilde{B}_n = DB_n D = (I - \Omega_n^2)^{-1}, \quad D = \text{diag}((-1)^{\lfloor i/2 \rfloor}), \quad D^2 = I,$$

where Ω_n is the *antisymmetric* tridiagonal matrix carrying the same off-diagonal entries $1, \dots, n-1$ that J_n carries symmetrically, so that $\Omega_n^2 = -DJ_n^2 D$. Cardinality statements and principal-minor statements — hence all of Theorem 2.3(iii) and Corollary 2.4 — are invariant under that conjugation, since $\det((DAD)_S) = \det(D_S)^2 \det(A_S)$. *Total nonnegativity is not*: it is $\tilde{B}_n$ that is the totally nonnegative min-kernel the deposited sampling algorithms exploit, while $B_n = (I + J_n^2)^{-1}$ carries alternating signs and is not. Accordingly, the released sampler is linked through $\tilde{B}_n = DB_n D$, not through B_n itself.

3 Classical structure and the new comparison

The polynomial family is classical; the new result begins with the adjacent-row coefficient comparison in §1. Reversed, the wall rows (7) are the rows of the triangle counting permutations by number of odd cycles: writing $T(n, k)$ for the number of permutations of $\{1, \dots, n\}$ with exactly k odd cycles, one has $[z^j]M_{n-1}(z) = T(n, n-2j)$. This is OEIS **A060524** [11], contributed in 2001, whose recorded row recurrence $t(n, x) = xt(n-1, x) + (n-1)^2 t(n-2, x)$ is (7) after reversal, and which the entry itself identifies as the Sheffer triangle for $(1/\cosh t, \tanh t)$.

Every link in the chain is in the literature. The combinatorial model for the coefficients is Viennot's [13], from Example 21 of chapter II of his 1983 lecture notes, by specializing the bijection between *histoires de Laguerre* and permutations. Heteyi [7] states it in the form used here: the monic Meixner polynomials of the second kind $M_n(x; \delta, \eta)$ (sans-serif, not the wall rows $M_n(z)$ of (7)) satisfy $b_n = (2n + \eta)\delta$ and $\lambda_n = (\delta^2 + 1)n(n + \eta - 1)$, so $(\delta, \eta) = (0, 1)$ gives precisely $b_n = 0$, $\lambda_n = n^2$, and “up to sign, the coefficient of x^j in $M_n(x; 0, 1)$ is the number of permutations of $\{1, 2, \dots, n\}$ having j odd cycles”; his Lemma 1.3 identifies the

moments of that family as the secant numbers E_{2n} . Mwafise and Barry [10] name the array on the generating-function side: the inverse of the exponential Riordan array $[\text{sech } x, \tanh x]$ is an orthogonal-polynomial coefficient array, they identify it as A060524 and as the odd-cycle count, and they print both $P_{n+1}(x) = xP_n(x) + n^2P_{n-1}(x)$ and its imaginary-argument twin $Q_{n+1}(x) = xQ_n(x) - n^2Q_{n-1}(x)$ obtained by $Q_n(x) = P_n(ix)/i^n$. The rotation is not cosmetic: $P_2(t) = t^2 + 1$ has nonreal zeros, so P_n itself is not an orthogonal polynomial sequence for a positive measure on the line; it is Q_n that is. Barry's earlier treatment [2] supplies the general mechanism by which a permutation statistic becomes an orthogonal-polynomial coefficient array with an LDU-factorized Hankel matrix of moments.

The measure and the continued fraction are older still. The secant numbers are the even moments of a sech density, $E_{2n} = \int_0^\infty y^{2n} \text{sech}(\pi y/2) dy$, and their generating function is the J-fraction with coefficients $\alpha_n = n^2$; as Sokal [12] records, these formulae were found by Stieltjes in 1889 and by Rogers in 1907, and given combinatorial proofs by Flajolet [6] in 1980. Deb and Sokal [5] reprint the same fraction with the same attribution. On the determinantal side, the cardinality theorem for a determinantal point process with Hermitian contraction kernel — that $|X|$ is a sum of independent Bernoulli variables with the eigenvalues as parameters — is textbook; see for instance Kulesza and Taskar [9].

Together these results give the one-row facts used here: real-rootedness, the Poisson-binomial law and Newton log-concavity. They do not control adjacent-row curvature in the coefficient index: root-variable orthogonality supplies interlacing, not signs of coefficient differences. Section 5 supplies that missing structure.

3.1 Neighbouring results

The nearest Turán inequalities are close in vocabulary and different in index. Krasikov [8] compares consecutive polynomial degrees pointwise in the root variable. Riordan-array log-concavity and the coefficientwise Hankel-total-positivity results of Zhu [14, 15, 16] concern one row or Hankel minors formed across rows. The S-fractions of Deb and Sokal [5] concern the continued-fraction side. None of these is a fixed-coefficient comparison of two adjacent rows through a Turán decrement.

The determinantal cardinality theorem itself is classical [9]. Here it identifies the Bernoulli parameters with the spectrum of the integer Jacobi section; Theorem 2.7 adds the explicit configuration map.

4 The algebraic reduction

4.1 Rows, ratios, and the level parameter

Throughout, F_r is the monic row (8) of degree $r - 1$ and $\Pi_r = F_r/(2r - 1)!$. Define the *factorially scaled row*

$$b_{r,k} = (2k + 1)! [u^k] F_r(u), \quad 0 \leq k \leq r - 1, \quad (13)$$

which is a positive integer, and the *scaled row ratios*

$$X_{r,k} = \frac{b_{r,k}}{b_{r,k-1}}, \quad 1 \leq k \leq r - 1. \quad (14)$$

Finally put

$$a_r = X_{r,1}, \quad (15)$$

the *level parameter* of the row.

The scaling is characterized by the shape row

$$c_k(a) = \frac{a^k}{(2k+1)!}, \quad (16)$$

for which every scaled ratio X_k is exactly a .

It is convenient to record how a_r is computed. Write $A_k(n) = [z^{\lceil n/2 \rceil - k}]M_n$ for the top reverse diagonals of the wall rows, so that $a_r = 6A_1(2r-2)/A_0(2r-2)$ by (13)–(15). These satisfy a closed triangular recurrence with strictly positive coefficients,

$$A_k(2m) = A_k(2m-1) + (2m)^2 A_k(2m-2), \quad A_k(2m+1) = A_{k-1}(2m) + (2m+1)^2 A_k(2m-1), \quad (17)$$

in which A_0 decouples, with $A_0(2m+1) = ((2m+1)!!)^2$.

4.2 Cell-local variables

Fix a cell (r, t) and let $F = (F_0, \dots, F_{r-1})$ be the row F_r normalized so that $F_{t-1} = 1$. Sans-serif letters denote quantities local to that cell. Put

$$x = \frac{F_t}{F_{t-1}}, \quad y = \frac{F_{t-2}}{F_{t-1}}, \quad z = \frac{F_{t+1}}{F_t}, \quad U = xy = \frac{F_t F_{t-2}}{F_{t-1}^2}, \quad W = \frac{z}{x} = \frac{F_{t+1} F_{t-1}}{F_t^2}, \quad (18)$$

so U and W are the two adjacent Newton ratios at the cell, and by (14),

$$x = \frac{X_{r,t}}{2t(2t+1)}, \quad U = \frac{(t-1)(2t-1)}{t(2t+1)} \cdot \frac{X_{r,t}}{X_{r,t-1}}, \quad W = \frac{t(2t+1)}{(t+1)(2t+3)} \cdot \frac{X_{r,t+1}}{X_{r,t}}. \quad (19)$$

The derived cell scalars are

$$\begin{aligned} \mathbf{s} &= (1-U) + x(1-UW) + x^2(1-W), & \mathbf{v} &= 1-U, \\ \mathbf{a} &= x(1-UW) + 2x^2(1-W), & \mathbf{d} &= x^2 + U + 2UWx, \\ C_1 &= 2\mathbf{s}(t-1) + \mathbf{a}, & C_2 &= \mathbf{s}(t-1)^2 + \mathbf{a}(t-1) + \mathbf{d}. \end{aligned} \quad (20)$$

Equivalently $\mathbf{s} = 1 + x + x^2 - xy - xz - xyz$ and $\mathbf{v} = 1 - xy$.

4.3 Lowering profiles

Let $m = r - 1$ and compare the row F_r with its predecessor through the *lowering profile*

$$\mu_k = \frac{F_k^-}{F_k} = 1 - \frac{L_k}{r}, \quad \ell = L_1, \quad \mu_k^0 = 1 - \frac{k\ell}{r}, \quad (21)$$

where the predecessor is scaled so that $F_0^- = F_0$, and μ^0 is the *linear profile* with the same first lowering. Here and below, $(\mu \odot F)_k = \mu_k F_k$ denotes pointwise multiplication, while $\Delta f_k = f_{k+1} - f_k$ is the forward difference in the coefficient index. The admissible profiles are constrained by

$$L_0 = 0, \quad L_m = r, \quad q_k = \Delta^2 L_k \geq 0 \quad (0 \leq k \leq m-2), \quad q_{k+1} \leq q_k \quad (0 \leq k \leq m-3), \quad (22)$$

that is, L is convex with decreasing curvature and pinned at both ends. Theorem 5.5 also assumes the natural orientation $\ell = L_1 \geq 0$; Lemma 4.2 proves it strictly for the adjacent-row profile. The point of the next lemma is that this profile really has the curvature hypotheses.

Lemma 4.1 (curvature of the adjacent-row profile). *The lowering profile defined by (21) satisfies (22).*

Proof. Put $p_{m,k} = [u^k]\Pi_{m+1}(u)$, $s_m = p_{m,0}$ and $\eta_k = p_{m-1,k}/p_{m,k}$, where $m = r - 1$. The chosen normalization gives the exact relation

$$\frac{L_k}{r} = 1 - \frac{s_m}{s_{m-1}}\eta_k.$$

The odd coefficient triangle is totally nonnegative by the planar-network argument of Lemma 5.1. Its monic rows $P_m = F_{m+1}$ also satisfy the positive same-column lowering

$$(m - u\partial_u)P_m(u) = \sum_{j=0}^{m-1} a_{m,j}P_j(u), \quad a_{m,j} = \frac{(2m+1)!(2m+2j+3)}{2(2j+1)!(2m-2j-1)(2m-2j+1)} > 0.$$

Write $\vartheta = u\partial_u$ and restrict coefficient-row determinants to the displayed consecutive columns. After positive column factors are removed,

$$\det(P_m, \vartheta P_m, P_{m-1})_{k:k+2} = c_{m,k}\Delta^2\eta_k, \quad \det(P_m, \vartheta P_m, \vartheta^2 P_m, P_{m-1})_{k:k+3} = 2d_{m,k}\Delta^3\eta_k,$$

with $c_{m,k}, d_{m,k} > 0$. Substituting the lowering identity in the first determinant turns it into the negative of a positive combination of minors of the coefficient triangle, and gives $\Delta^2\eta_k \leq 0$. In the four-column expansion, paired terms carry an adjacent difference of connection ratios. Clearing its positive denominator leaves 36 strictly positive integer coefficients, so $\Delta^3\eta_k \geq 0$.

The displayed relation makes L_k/r a decreasing affine function of η_k . Therefore $\Delta^2 L_k \geq 0$ and $\Delta^3 L_k \leq 0$, the latter being exactly $q_{k+1} \leq q_k$. The endpoint coefficients give $L_0 = 0$ and $L_m = r$. $\square$

Lemma 4.2 (first lowering). *With the convention $a_1 = 0$, every $r \geq 2$ satisfies*

$$\ell = \frac{r(a_r - a_{r-1})}{a_r}, \quad [r(a_r - a_{r-1})]^2 - a_r > \frac{57}{20}.$$

Consequently

$$\frac{\sqrt{a_r + 57/20}}{a_r} < \ell \leq \frac{r}{r-1}. \quad (23)$$

Proof. Since $a_r = 6F_{r,1}/F_{r,0}$ and the predecessor is normalized to have the same constant coefficient, the formula for $\ell = L_1$ is immediate from (21). For the strict estimate put $m = r - 1$, $E_m = a_r/6$, $c_m = E_m - E_{m-1}$, $b_m = (m+1)c_m$ and $H_m = 6b_m^2 - E_m$. Then the displayed defect is $6H_m$.

For the tail, put

$$h_n = \left(\frac{(2n)!!}{(2n-1)!!} \right)^2, \quad T_n = 1 + \sum_{j=1}^n h_j^{-1}, \quad R_n = h_n T_n,$$

and

$$w_n = \frac{h_n T_n^2}{(2n+1)^2}, \quad \mathcal{P}_n = \sum_{i=0}^n w_i, \quad g_n = \frac{h_n}{2n+1}, \quad \mathcal{F}_n = g_n^2 T_n^3, \quad \mathcal{C}_n = \mathcal{P}_n - \mathcal{F}_n/3.$$

Writing $\theta = (T_n - T_{n-1})/T_{n-1} \leq 1/(4n)$, direct cancellation gives

$$\frac{\mathcal{F}_n - \mathcal{F}_{n-1} - 3w_n}{g_n^2 T_{n-1}^3} = \frac{8n^2 - 1}{16n^4} - \theta^2(3 + 2\theta) \geq \frac{10n^2 - n - 2}{32n^4} > 0.$$

Thus $\mathcal{C}_n$ decreases. The exact seed $\mathcal{C}_{1000} < -1/5$ implies, for $m \geq 1001$,

$$0 \leq b_m \leq B_m = \frac{m+1}{R_m} \left[\frac{(2m-1)^2(R_m-1)^2}{48m^4} - \frac{h_m}{5(R_m-1)} \right].$$

The elementary bounds $R_m \geq 4m+1$, $3m < h_m < 4m$ and $R_m \leq 2m^{5/4}$ give $B_m \geq 3/16$. With

$$\mathcal{D} = (2m+1)^2 + 4(m+1)^2 R_m, \quad \alpha = \frac{(m+2)(R_m-1)(2m+1)^2}{(m+1)\mathcal{D}}, \quad \beta = \frac{(m+2)R_m}{\mathcal{D}},$$

the recurrence is $b_{m+1} = \alpha b_m + \beta$, with $0 < \alpha < 1$. If $d = \beta - (1-\alpha)B_m$, clearing denominators reduces $d \geq 1/(12m)$ to a quartic, which the bounds above put above $32m^5(6m^3-5) > 0$. Since $b_1 = 2/5$ and $\alpha, \beta > 0$, the recurrence also gives $b_m > 0$ on every row. Thus $c_m > 0$, so $a_r > a_{r-1}$ and $\ell > 0$; this orients the square root in (23).

Finally

$$H_{m+1} - H_m = 6[(\alpha b_m + \beta)^2 - b_m^2] - \frac{\alpha b_m + \beta}{m+2}.$$

As a function of b_m , the right-hand side is concave on $[0, B_m]$; it is positive at 0, and at B_m it is at least $[48mB_m + 2 - m]/[24m^2(m+2)] > 0$. Hence H_m increases for $m \geq 1001$. Exact rational arithmetic verifies $H_m > 19/40$ for $1 \leq m \leq 1001$ (the minimum is at $m = 143$), proving the strict estimate on every row. Finally, convexity of L , together with $L_0 = 0$ and $L_{r-1} = r$, puts its first slope below the average slope and gives $\ell \leq r/(r-1)$. $\square$

4.4 A cross-row consequence of the profile curvature

The curvature statements just proved say more about the array than the square-tail argument consumes. The following is immediate from them and is recorded because it is the clean cross-row comparison in the coefficient index.

Corollary 4.3 (cross-row ratio). *Let $\mu_k = 1 - L_k/r$ be the lowering profile (21). For $1 \leq k \leq r-2$,*

$$\frac{X_{r-1,k}}{X_{r,k}} = \frac{\mu_k}{\mu_{k-1}},$$

the array is strictly increasing across rows, $X_{r,k} > X_{r-1,k}$, and $k \mapsto X_{r,k}/X_{r-1,k}$ is strictly increasing. Equivalently, the adjacent row-index minors of the X -array have a strict sign.

Proof. By Lemma 4.1 and Lemma 4.2, $L_0 = 0$, $L_1 = \ell > 0$ and $\Delta^2 L_k \geq 0$, so L is strictly increasing and μ is strictly decreasing and positive on the stated range; the displayed identity is (21) read as a quotient, and $X_{r,k} > X_{r-1,k}$ follows. For the monotone quotient write $M = \mu_{k-1}$, let $p = (L_k - L_{k-1})/r > 0$ be the current increment and $p+q$ with $q \geq 0$ the next, so that $\mu_k = M - p$ and $\mu_{k+1} = M - 2p - q$. Then

$$\mu_k^2 - \mu_{k-1}\mu_{k+1} = p^2 + Mq > 0,$$

so μ is strictly log-concave and μ_{k-1}/μ_k strictly increases. $\square$

The hypothesis is the proved curvature of the lowering profile, not a total positivity assertion about the connection kernels of the derivative and lowering identities: those kernels are *not* totally positive, and their exact 2×2 minors occur with both signs. The usable positivity lives in the coefficient triangle and in the profile curvature it induces.

4.5 Symbol table

symbol	meaning
$K_r, o(\pi)$	odd-cycle statistic $(o(\pi) - 1)/2$; number of odd cycles
$M_n(z)$	wall rows (7)
F_r, Π_r	monic row (8); its normalization $F_r/(2r - 1)!$
J_n, Ω_n	symmetric sech Jacobi matrix; its antisymmetric twin
D_r, G_r	parity incidence matrix (5); Gram matrix $D_r^\top D_r$
$\xi_{r,j}, \theta_{r,j}$	positive eigenvalues of J_{2r-1} ; $1/(1 + \xi_{r,j}^2)$
$B_n, \tilde{B}_n, \mathcal{K}_r$	$(I + J_n^2)^{-1}$; its conjugate DB_nD ; parity kernel $(I + G_r)^{-1}$
$L(f)_t$	coefficientwise Turán operator (1)
$S_{r,t}, V_{r,t}, d_{r,t}, \rho$	the derived quantities (2), (3)
$b_{r,k}, X_{r,k}, a_r$	scaled row (13); ratios (14); level parameter $X_{r,1}$
Δ_r	first decrement $X_{r,1} - X_{r,2}$
x, y, z, U, W	cell-local ratios (18)
$\mathbf{s}, \mathbf{v}, \mathbf{a}, \mathbf{d}, C_1, C_2$	cell-local scalars (20)
L_k, μ_k, ℓ	lowering profile (21)
$\sigma, \sigma_{\text{lin}}$	predecessor-row analogue of $\mathbf{s}$; its linear-profile value
$A_k(n), R_m$	top reverse diagonals (17); normalizer ladder (30)
ε, ψ	normalizer scalar; normalized cross-row surplus (32)

Three symbols in §3 belong to the cited sources and are used nowhere else: P_n, Q_n for the Mwafise–Barry pair, E_{2n} for the secant numbers, and $M_n(x; \delta, \eta)$ for the monic Meixner polynomials of the second kind. That last one is written with a sans-serif M throughout precisely because it is *not* the wall row $M_n(z)$ of (7); the two carry the same letter in their respective literatures and are different objects.

5 Analytic inequalities on the ratio layer

5.1 Monotonicity and convexity of the scaled ratios

Lemma 5.1 (total positivity in the coefficient index). *The triangular array $([u^k]F_r)_{r \geq 1, k \geq 0}$ is totally nonnegative, and so is the column-scaled array $B = (b_{r,k})$ of (13).*

Proof. The odd-cycle array is the odd parity block of the exponential Riordan array $[(1 - x^2)^{-1/2}, \operatorname{artanh} x]$. Its inverse, after the checkerboard signs are removed, is the odd block of $[\sec x, \tan x]$, which is the path matrix of an acyclic planar network in which up-steps carry weight 1 and a down-step from height h carries weight h^2 . By the Lindström lemma that inverse block is totally nonnegative, and Jacobi’s complementary-minor identity together with the two checkerboard sign matrices makes every minor of the original odd block nonnegative. Multiplying column k by $(2k + 1)! > 0$ preserves the sign of every minor. $\square$

Theorem 5.2 (ratio signs). *For every $r \geq 3$,*

$$X_{r,k-1} \geq X_{r,k} \quad (2 \leq k \leq r - 1), \quad X_{r,k-1} - 2X_{r,k} + X_{r,k+1} \geq 0 \quad (2 \leq k \leq r - 2).$$

Proof. Two ingredients. The first is Lemma 5.1: the array $B = (b_{r,k})$ is totally nonnegative.

Second, two exact lowering identities connect a row to the rows strictly below it. Twice differentiating the bivariate generating function of $A_n(y)$ multiplies it by $\operatorname{artanh}(x)^2$; writing

$H_d^o = \sum_{i=0}^{d-1} 1/(2i+1)$ and using $[x^{2d}] \operatorname{artanh}(x)^2 = H_d^o/d$, coefficient extraction gives the *positive lowering*

$$b_{r,k+1} = \sum_{j=1}^{r-1} \delta_{r,j} b_{j,k}, \quad \delta_{r,j} = \frac{(2r-1)!}{(2j-1)!} \cdot \frac{H_{r-j}^o}{r-j}, \quad (24)$$

and the second lowering, unchanged by the column scaling, is

$$(r-1-k) b_{r,k} = \sum_{j=1}^{r-1} \ell_{r,j} b_{j,k}, \quad \ell_{r,j} = \frac{(2r-1)! (2r+2j-1)}{2(2j-1)! (2r-2j-1)(2r-2j+1)}. \quad (25)$$

Both connection sequences are strictly positive.

Let Sb_r denote the shifted row $(Sb_r)_k = b_{r,k+1}$. By (24) it is a positive combination of rows strictly below r , so total nonnegativity forces, for $0 \leq k \leq r-3$, the 2×2 minor $\det \begin{pmatrix} b_{r,k} & b_{r,k+1} \\ (Sb_r)_k & (Sb_r)_{k+1} \end{pmatrix} \leq 0$. Dividing by $b_{r,k} b_{r,k+1} > 0$ and reindexing gives the first sign.

For convexity, let $Kb_r = k b_{r,k}$. On three consecutive columns, for $0 \leq k \leq r-4$,

$$\det(b_r, Kb_r, Sb_r) = b_{r,k} b_{r,k+1} b_{r,k+2} [X_{r,k+3} - 2X_{r,k+2} + X_{r,k+1}].$$

Let $(Lb_r)_k = \sum_{j < r} \ell_{r,j} b_{j,k}$, so (25) gives $Kb_r = (r-1)b_r - Lb_r$; the repeated row drops out, and expanding $-\det(b_r, Lb_r, Sb_r)$ and pairing indices $i < j$ leaves a sum of terms $[\ell_{r,j} \delta_{r,i} - \ell_{r,i} \delta_{r,j}] \det(b_i, b_j, b_r)$. Each minor is nonnegative by total nonnegativity, so it suffices to sign the bracket, that is, to show $\delta_{r,j}/\ell_{r,j}$ is decreasing in j at fixed r . Direct cancellation in (24)–(25) gives, with $d = r-j$,

$$\frac{\delta_{r,j}}{\ell_{r,j}} = \frac{2(4d^2-1)H_d^o}{d(4r-2d-1)}. \quad (26)$$

Writing $H = H_d^o$ and $s = j-1 \geq 0$, so that $r = d+s+1$, and using $H_{d+1}^o = H_d^o + 1/(2d+1)$, the increment of (26) in d is

$$\frac{2[N_1 + N_2]}{d(d+1)(2d+4s+1)(2d+4s+3)},$$

where

$$N_1 = H(16d^3 + 16d^2s + 28d^2 + 16ds + 12d + 4s + 1), \quad N_2 = 4d^3 + 8d^2s + 12d^2 + 12ds + 9d,$$

whose twelve numerator coefficients are all strictly positive. So (26) increases with d and hence decreases with j , every bracket is nonnegative, and the 3×3 determinant is nonnegative. Reindexing $k+2$ as k gives the second assertion. $\square$

Corollary 5.3 (corridor). *Put $\Delta_r = X_{r,1} - X_{r,2}$. For every $r \geq 3$,*

$$0 \leq X_{r,k-1} - X_{r,k} \leq \Delta_r \quad (2 \leq k \leq r-1), \quad a_r - (k-1)\Delta_r \leq X_{r,k} \leq a_r \quad (1 \leq k \leq r-1).$$

Proof. Convexity makes the first differences $X_{r,k-1} - X_{r,k}$ nonincreasing in k and monotonicity makes them nonnegative, so each is bounded by the first one, Δ_r . Summing gives the second display. $\square$

Corollary 5.3 reduces every decrement on a row to the single level-2 quantity Δ_r . That is what the next theorem bounds.

5.2 The global decrement bound

Theorem 5.4 (decrement bound). *For every $r \geq 3$, $0 \leq \Delta_r < 5$. Consequently, throughout every row,*

$$0 \leq X_{r,k-1} - X_{r,k} < 5, \quad a_r - 5(k-1) < X_{r,k} \leq a_r. \quad (27)$$

Proof. The lower sign is Theorem 5.2. For the upper sign put $m = r - 1$ and retain h_j, T_j from Lemma 4.2. Define the triangular Wallis ladder by

$$\begin{aligned} B_0(j) &= T_j, & C_0(j) &= 1, \\ B_k(j) &= B_k(j-1) + h_j^{-1} C_k(j-1), & C_k(j) &= C_k(j-1) + \frac{h_j}{(2j+1)^2} B_{k-1}(j), \end{aligned}$$

for $j, k \geq 1$, with $B_k(0) = 0$ for $k \geq 1$, $C_1(0) = 1$ and $C_k(0) = 0$ for $k \geq 2$. Exact reversal of (17) gives

$$E_k := \frac{B_k(m)}{T_m} = \frac{[u^k] \Pi_r}{[u^0] \Pi_r}, \quad \Delta_r = 6E_1 - 20 \frac{E_2}{E_1}.$$

Thus, with $\beta_k = (\pi/2) B_k(m)$, multiplying $5 - \Delta_r$ by the positive $\beta_0 \beta_1$ clears denominators to

$$H = 5\beta_0 \beta_1 - 6\beta_1^2 + 20\beta_0 \beta_2, \quad (28)$$

so $H > 0$ is equivalent to $\Delta_r < 5$. The finite leg reads the first three integer coefficients off the row recurrence

$$F_{r+1} = (u + 8r^2 - 4r + 1)F_r - [2(r-1)(2r-1)]^2 F_{r-1}, \quad (29)$$

and, writing $p_i = [u^i] F_r$, checks the cleared numerator $5p_0 p_1 - 6p_1^2 + 20p_0 p_2 > 0$ exactly at every row through $r = 1001$. The infinite leg uses rational polynomial envelopes $\beta_k^{\text{lo}} \leq \beta_k \leq \beta_k^{\text{hi}}$ in the nonnegative shift $s = \tau_m - \tau_{1000}^{\text{lo}}$ of the Wallis clock $\tau_m = \beta_0 = (\pi/2) T_m$, anchored at $r = 1001$. Since every lower-envelope coefficient is nonnegative, a valid lower bound for H is $H_{\text{lo}} = 5\beta_0 \beta_1^{\text{lo}} - 6(\beta_1^{\text{lo}})^2 + 20\beta_0 \beta_2^{\text{lo}}$, which after exact collection has degree six with every rational coefficient nonnegative and strictly positive constant term. Hence $H > 0$ for all $r \geq 1001$. The two legs overlap at $r = 1001$, so their union is gap-free. $\square$

5.3 Linear extremality and the scalar reduction

Theorem 5.5 (linear profiles are extremal). *Let F be the current row and let μ be any lowering profile satisfying (22) and $\ell = L_1 \geq 0$, with μ^0 the linear profile carrying that same first lowering. Then for $3 \leq t \leq m - 1$,*

$$L_t((1+u)(\mu \odot F)) \leq L_t((1+u)(\mu^0 \odot F)).$$

That is, the predecessor Turán determinant is maximized by the linear profile.

Proof. Put $H_k = L_k - k\ell$ and $p_k = \Delta H_k$, so $H_0 = H_1 = p_0 = 0$, $\Delta p_k = q_k$, and p is increasing and concave. For $k \geq 2$, $H_k^2 - H_{k-1} H_{k+1} = p_{k-1}^2 - H_{k-1} q_{k-1}$, and because q is decreasing, $p_{k-1} \geq (k-1)q_{k-1}$ and $H_{k-1} \leq (k-2)p_{k-1}$, making the right side nonnegative. So the pure defect H is log-concave; F is log-concave; the pointwise product $H \odot F$ is log-concave; and convolution with $(1, 1)$ preserves the three-term inequality, giving $L_t((1+u)(H \odot F)) \geq 0$.

For the mixed term, set $k = t - 2$, $n = m - k$, $d_k = \Delta L_k$ and normalize by $M = \mu_k > 0$. Put

$$a = \frac{\ell}{rM}, \quad p = \frac{d_k - \ell}{rM}, \quad h = \frac{H_k}{rM}, \quad q = \frac{q_k}{rM}, \quad v = \frac{q_{k+1}}{rM},$$

and define

$$G = (1 + u)((\mu/M) \odot F), \quad J = \frac{1}{rM}(1 + u)(H \odot F).$$

Since $\mu^0 = \mu + H/r$, we have $G + J = (1 + u)((\mu^0/M) \odot F)$. If $p = 0$, the local defect variables vanish and the comparison is immediate. For $p > 0$, the constraints (22) together with the endpoint budget cut out an exact five-dimensional polytope in (a, p, h, q, v) . With $d = a + p$, its inequalities are

$$0 \leq p \leq d, \quad 0 \leq q \leq \frac{p}{k}, \quad 0 \leq v \leq q, \quad \frac{k-1}{2}p \leq h \leq (k-1)p, \quad nd + (n-1)q + (n-2)v \leq 1.$$

After substituting the degree-aware Newton bounds (34) in the safe direction, the mixed polarization $C = 2G_t J_t - G_{t-1} J_{t+1} - J_{t-1} G_{t+1}$ becomes $C = \Gamma_0 + \Gamma_1 x + \Gamma_2 x^2$ with each Γ_i/p having positive denominator. After that denominator is cleared, its numerator has a tensor-product Bernstein expansion on the unit cube, all of whose coefficients are nonzero polynomials in (k, n) with nonnegative integer coefficients. The polarization identity $L_t(G + J) - L_t(G) = C + L_t(J)$ then gives the theorem. $\square$

Theorem 5.5 converts the row comparison into a scalar one. Three further cell quantities are needed, and we define them before use.

First, on the current normalization $F_{t-1} = 1$, put

$$\sigma = L_t((1 + u)(\mu \odot F)), \quad \sigma_{\text{lin}} = L_t((1 + u)(\mu^0 \odot F)).$$

No second normalization is applied to the lowered rows. Thus Theorem 5.5 is exactly $\sigma \leq \sigma_{\text{lin}}$.

Second, the normalizer ladder used in the proof of Lemma 4.2 is exactly the top reverse-diagonal ratio $R_m = A_0(2m)/A_0(2m-1) = h_m T_m$. By (17) it satisfies

$$R_1 = 5, \quad R_m = 1 + \left(\frac{2m}{2m-1}\right)^2 R_{m-1}, \quad 4m+1 \leq R_m < 4m(m+1) + 1. \quad (30)$$

Both bounds follow by induction; after inserting the upper bound at $m-1$ and clearing $(2m-1)^2$, the unused margin in the upper step is exactly $4m$. The *normalizer scalar* of the cell is $\varepsilon = \beta_r(2 - \beta_r)$ with $\beta_r = b(1+A) - A$, $A = 1/[4r(r-1)]$ and $b = 1/R_{r-1}$. Equation (30) gives

$$\frac{1}{4r(r-1)+1} < b \leq \frac{1}{4r-3} < 1,$$

so $0 < \beta_r < b < 1$. Consequently

$$0 < \varepsilon < \frac{2}{4r-3}. \quad (31)$$

Third, put

$$\psi = \varepsilon + (1 - \varepsilon) \frac{\mathbf{s} - \sigma}{\mathbf{s}}, \quad (32)$$

the normalized cross-row surplus of the cell. Here is the exact bridge back to the original ratio. Let $c = [u^{t-1}] \Pi_r > 0$, so that $\Pi_r = cF$ on the current normalization. Then $S_{r,t} = c^2 \mathbf{s}$ and $V_{r,t} = c^2 \mathbf{v}$. The normalizers match exactly through

$$1 - \varepsilon = (1 - \beta_r)^2 = \left[\frac{2r-1}{2r} \frac{\Pi_{r-1}(0)}{\Pi_r(0)} \right]^2, \quad c^2 \sigma = \left[\frac{\Pi_r(0)}{\Pi_{r-1}(0)} \right]^2 S_{r-1,t}.$$

Thus the weighted predecessor term in $d_{r,t}$ is $c^2(1 - \varepsilon)\sigma$. Hence

$$d_{r,t} = c^2 [\mathbf{s} - (1 - \varepsilon)\sigma] = c^2 \mathbf{s} \psi, \quad \rho(r, t) = \frac{(r\psi)^2 \mathbf{s}}{2\mathbf{v}}. \quad (33)$$

Thus Theorem 1.1 at the cell follows if $r\psi \geq \sqrt{2\mathbf{v}/\mathbf{s}}$. (After Lemma 6.6, the condition is also necessary.) With $n = r - t + 1$, the degree-aware Newton bounds are

$$U \leq \frac{(t-1)(n-1)}{tn}, \quad W \leq \frac{t(n-2)}{(t+1)(n-1)}. \quad (34)$$

In particular $U, W \leq 1$, so $\mathbf{s} - \mathbf{v} = x(1 - UW) + x^2(1 - W) \geq 0$ and therefore $\mathbf{v} \leq \mathbf{s}$; hence

$$r\psi \geq \sqrt{2} \implies \text{Theorem 1.1 at } (r, t), \quad (35)$$

a sufficient condition uniform in the cell.

Now direct expansion for the linear profile gives $\mathbf{s} - \sigma_{\text{lin}} = C_1\ell/r - C_2\ell^2/r^2$ with C_1, C_2 as in (20) — note that the linear and quadratic pieces never separate into independent margins. The substitution from Lemma 4.2 is safe for a concrete reason. On the interior range used below, $t, n \geq 3$, and the degree-aware Newton bounds reduce $mC_1 - 2C_2 \geq 0$ to the following three positive coefficients of $1, x, x^2$:

$$\frac{2(n-1)(t-1)(n+t-2)}{nt}, \quad \frac{2(n+t-2)(2nt-n-3t+3)}{n(t+1)}, \quad \frac{2t(n-2)(n+t-2)}{(n-1)(t+1)}.$$

Therefore $C_1\ell - C_2\ell^2/r$ is increasing for $0 \leq \ell \leq r/m$. Writing $L_1^{\text{lo}} = \sqrt{a_r + 57/20}/a_r < \ell$, the obligation $\rho \geq 1$ is implied by

$$r\varepsilon\mathbf{s} + (1 - \varepsilon) \left[C_1 L_1^{\text{lo}} - C_2 (L_1^{\text{lo}})^2 / r \right] \geq \sqrt{2\mathbf{s}\mathbf{v}}, \quad (36)$$

which is (32) cleared of its denominator. Whenever (36) holds at a cell, so does Theorem 1.1 there.

The same algebra supplies a row-independent upper comparison. With $k = t - 2$, degree-aware Newton gives

$$\sigma \leq \sigma_{\text{lin}} \leq [1 - k\ell/r]^2 \mathbf{s}. \quad (37)$$

Monotonicity on $0 \leq \ell/r \leq 1/m$ reduces to three positive Newton coefficients. Substituting into (32) yields

$$r\psi \geq r\varepsilon + (1 - \varepsilon) \left[2q - \frac{q^2}{r} \right], \quad q = (t-2) \frac{\sqrt{a_r + 57/20}}{a_r}, \quad (38)$$

so by (35) every cell whose right-hand side reaches $\sqrt{2}$ satisfies Theorem 1.1.

6 Covering the domain

Four regions cover the interior $r \geq 4$, $2 \leq t \leq r - 2$. Two terminal coefficient identities then complete the natural domain of Theorem 1.1.

6.1 Fixed columns

Proposition 6.1 (seven complete columns). *For each $t \in \{2, 3, 4, 5, 6, 7, 8\}$ there is an explicit threshold $R_0(t)$ with $\rho(r, t) \geq 1$ for every $r \geq R_0(t)$, namely*

$$\begin{aligned} R_0(2) &= 313, & R_0(3) &= 116, & R_0(4) &= 250, & R_0(5) &= 700, \\ R_0(6) &= 2004, & R_0(7) &= 6200, & R_0(8) &= 20000. \end{aligned}$$

The finite core through $r = 503$ reaches the tails for $t = 2, 3, 4$. One exact-integer stream up to the displayed anchors supplies the remaining gaps for $t = 5, 6, 7, 8$. Hence $\rho(r, t) \geq 1$ at every interior cell of columns $t = 2, \dots, 8$; their two terminal cells are handled separately in Proposition 6.4.

These thresholds are certificate anchors, not claimed minima. Fixed-column tails cannot cover t growing with r ; that is the role of the joint window. The seven tail polynomials and the 26,888 prefix cells are in the replay packet.

6.2 The joint window

Theorem 6.2 (uniform bound on the joint window). *Suppose $2 \leq t \leq r - 2$ and*

$$r \geq 504, \quad t \geq 6, \quad a_r \geq 5t, \quad t < \sqrt{a_r} + 2. \quad (39)$$

Then (36) holds, and hence so does Theorem 1.1.

Proof. The proof uses Theorems 5.2 and 5.4 only through the corridor (27), and no lower bound on any adjacent-difference margin. Write $Y = X_{r,t-1}$, $B = X_{r,t}$, $C = X_{r,t+1}$ and $\alpha = (t-1)(2t-1)/[t(2t+1)]$, $\beta = t(2t+1)/[(t+1)(2t+3)]$, so that (19) reads $x = B/[2t(2t+1)]$, $U = \alpha B/Y$, $W = \beta C/B$. The corridor and $a_r \geq 5t$ give $Y \geq a_r - 5(t-2) \geq 10$, $B \geq Y - 5$, $C \leq B$ and $a_r \leq Y + 5(t-2)$, hence $x \geq x_0 = (Y-5)/[2t(2t+1)]$, $U \geq u_0 = \alpha(1-5/Y)$ and $W \leq \beta$. With $R = s/v$ the identity $R = 1 + x + (xU + x^2)(1-W)/(1-U)$ gives $R \geq R_{\min} := 1 + x_0 + (x_0 u_0 + x_0^2)(1-\beta)/(1-u_0)$. The three positive summands of s carry weights $2(t-1)$, $2t-1$ and $2t$ in C_1 , so $C_1 \geq v[(2t-1)R-1]$ and $C_1^2/(2a_r s v) \geq [(2t-1)R-1]^2/(2a_r R)$, an expression increasing in $R \geq 1$ and decreasing in a_r . Substituting the two extremes yields the core inequality

$$C_1^2 \geq \frac{101}{100} \cdot 2a_r s v \quad (t \geq 6, Y \geq 10). \quad (40)$$

After clearing the positive denominator, the difference in (40) is a polynomial of bidegree (6, 14) in (Q, T) where $Y = Q + 10$ and $t = T + 6$. For $t \geq 8$ set $T = S + 2$; all 74 monomial coefficients in (Q, S) are strictly positive. At $t = 6, 7$, namely $T = 0, 1$, two exact degree-six Sturm sequences have equal sign-variation counts at 0 and $+\infty$ with positive value at zero, so neither has a nonnegative root.

The quadratic term stays correlated: $(t-1)C_1 - C_2 = s(t-1)^2 - d$, and the degree-aware Newton bounds $U \leq (t-1)/t$, $W \leq t/(t+1)$, $UW \leq (t-1)/(t+1)$ make the three coefficients in powers of x equal to $(t-1)(t-2)/t$, $2(t-1)(t-2)/(t+1)$ and $t(t-3)/(t+1)$, all nonnegative for $t \geq 3$; hence $C_2 \leq (t-1)C_1$ and $C_1 L_1^{\text{lo}} - C_2 (L_1^{\text{lo}})^2/r \geq (C_1/\sqrt{a_r})[1 - (t-1)/(r\sqrt{a_r})]$. On the window (39) we have $a_r \geq 5t \geq 30$ and $t < \sqrt{a_r} + 2$, so $1 - (t-1)/(r\sqrt{a_r}) > 1 - 2/r$. Dropping the positive $r\varepsilon s$ term in (36) and combining (40), the last display and (31), it remains to check $\frac{101}{100}[1 - 2/(4r-3)]^2[1 - 2/r]^2 \geq 1$. Both factors increase with r ; at $r = 504$ the unsquared product is $504761/507276$ and the exact squared surplus is $256361621/25732894017600 > 0$. $\square$

6.3 The top region

Theorem 6.3 (top region). *Every interior cell with*

$$r \geq 4, \quad 2 \leq t \leq r - 2, \quad t \geq \sqrt{a_r} + 2 \quad (41)$$

satisfies Theorem 1.1.

Proof. The threshold gives $q \geq 1$ in (38); also $L_1^{\text{lo}} < \ell \leq r/(r-1)$ and $t \leq r-2$, so $q < r$. The quadratic $2q - q^2/r$ is concave, so on $[1, r]$ it attains its minimum at an endpoint and is therefore at least $2 - 1/r$. With (31),

$$r\psi > \left[1 - \frac{2}{4r-3}\right] \left(2 - \frac{1}{r}\right),$$

and both factors increase with r . At $r = 4$ the right side equals $77/52$, and $(77/52)^2 - 2 = 521/2704 > 0$, so $r\psi > \sqrt{2}$, which is the threshold in (38). $\square$

6.4 The finite core

An exact definition-level witness checks every interior cell on

$$4 \leq r \leq 503, \quad 2 \leq t \leq r - 2, \quad (42)$$

which contains $\sum_{r=4}^{503} (r - 3) = 125,250$ cells. The cancelled integer expression for ρ is the following. Write $\widehat{S}_{r,t} = L((1+u)F_r)_t$ and $\widehat{V}_{r,t} = L(F_r)_{t-1}$ for the Turán forms of the reversed integer rows $F_r(u) = u^{r-1}M_{2r-2}(1/u) = (2r-1)!\Pi_r(u)$, with $\widehat{S}_{r-1,t}$ built from F_{r-1} in the same way, and

$$\mathcal{N}_{r,t} = r^2 \widehat{S}_{r,t} - (2r-1)^4 (r-1)^2 \widehat{S}_{r-1,t}, \quad \rho(r,t) = \frac{\mathcal{N}_{r,t}^2}{2r^2 \widehat{S}_{r,t} \widehat{V}_{r,t}}. \quad (43)$$

All three are integers, and (43) clears the factorials that (2) carries; $\mathcal{N}_{r,t}$ is not $d_{r,t}$, and $\widehat{S}, \widehat{V}$ are not S, V . The identity (43) is checked against the uncanceled definition (2) on every cell through $r = 18$; every one of the 125,250 comparisons is then strictly positive.

6.5 The two terminal cells

Proposition 6.4 (upper edge). *For every integer $r \geq 2$,*

$$\rho(r,r) = \frac{r^2}{2}.$$

For every integer $r \geq 3$,

$$\rho(r,r-1) > \frac{2r^2}{9} \geq 2.$$

Proof. Put $n = 2r - 2$, $a = [z]M_n$, $b = [z^2]M_n$ and $c = 1/(2r-1)!$. By (8), the top three coefficients of Π_r are c, ac, bc . At $t = r - 1$, direct substitution into (2) gives

$$S_{r,r-1} = c^2 D, \quad V_{r,r-1} = c^2 E, \quad d_{r,r-1} = c^2 (D - K),$$

where

$$D = a^2 + a + 1 - b, \quad E = a^2 - b, \quad K = \frac{(2r-1)^4 (2r-2)^2}{4r^2}.$$

Strict Newton gives $E > 0$, and $D = E + a + 1 > E$. Using the two formulas of Lemma 2.2 and writing $X = r - 3$, exact collection gives

$$D - 3K = \frac{R(X)}{90r^2},$$

with

$$\begin{aligned} R(X) = & 320X^8 + 7104X^7 + 62924X^6 + 295836X^5 + 808325X^4 \\ & + 1294641X^3 + 1133721X^2 + 425439X + 7020. \end{aligned}$$

Every coefficient is positive. Thus $D - K > 2D/3$, and

$$\rho(r,r-1) = \frac{r^2(D-K)^2}{2DE} > \frac{2r^2 D}{9E} > \frac{2r^2}{9} \geq 2.$$

At $t = r$, the top coefficient gives $S_{r,r} = V_{r,r} = c^2$, while $S_{r-1,r} = 0$. Hence $d_{r,r} = c^2$ and $\rho(r,r) = r^2/2$. $\square$

6.6 The optimal uniform constant

Theorem 1.2 is proved by partitioning the domain into a region where ρ is bounded below by the rational barrier $203/200$ and the finite core, which is the only region that can approach the minimum. The barrier is chosen because $\rho(129, 2) < 203/200$ strictly, with surplus $203/200 - \rho(129, 2) = 4.2383 \dots \times 10^{-4}$.

- (i) *Terminal columns.* $\rho(r, r) = r^2/2$ and $\rho(r, r-1) > 2r^2/9$ are Proposition 6.4; both exceed 2 for $r \geq 2$, $r \geq 3$.
- (ii) *Fixed columns* $t = 2, 3, 4, 5$. The tail certificates are retargeted from $\rho \geq 1$ to $\rho \geq 203/200$, re-anchoring the four columns at $m_0 = 503, 503, 503, 1000$. For $t = 2$ and $t = 5$ the move is forced: the old anchors 312 and 699 lack coefficient capacity for the larger constant.
- (iii) *Columns* $t = 6, 7, 8$. The monotone level parameter a_r crosses 25, 30, 35, 36, 40 first at rows 1468, 3634, 8414, 9884, 18457. Those crossings send the unbounded parts of these columns into the top region (Theorem 6.3, floor 5929/5408) or the joint window. On the joint window the core inequality (40) holds with $513/500$ in place of $101/100$ on the same domain — certified by the same Sturm construction in the packet's reduction component, and close to sharp, since $103/100$ already fails — so the row-corrected floor at the worst row $r = 504$ is $(513/500)(504761/507276)^2 = 4840889675299/4765350744000 > 203/200$, leaving three bounded windows $504 \leq r \leq 3633$, $1468 \leq r \leq 8413$ and $9884 \leq r \leq 18456$, of $3130 + 6946 + 8573$ cells; with the 497 residual $t = 5$ cells this is a finite stream of 19,146 exact comparisons.
- (iv) *Columns* $t \geq 9$. If a cell is not in the top region then $a_r > (t-2)^2 > 5t$, by $(t-2)^2 - 5t = (t-9)^2 + 9(t-9) + 4 > 0$, so it lies in the joint window. The dichotomy is exhaustive.
- (v) *The finite core.* On $4 \leq r \leq 503$, $2 \leq t \leq r-2$ every one of the 125,250 cells is compared with $\rho(129, 2)$ by exact cross-multiplication of (43). Exactly one cell lies at or below it, namely $(129, 2)$ itself, where equality holds.

Steps (i)–(iv) give $\rho > 203/200 > \rho(129, 2)$ off the core, and step (v) gives $\rho > \rho(129, 2)$ on the core away from $(129, 2)$. Hence the minimum is $\rho(129, 2)$, attained only there, which is Theorem 1.2.

Two diagnostics accompanied step (v), used nowhere in the proof: the 500 column minima of the core increase strictly in t , and the $t = 2$ column is unimodal on the core, decreasing for $5 \leq r \leq 129$ and increasing for $130 \leq r \leq 503$. Both are statements about the core only. Only 32 of the 125,250 core cells lie below $203/200$, all of them in column $t = 2$ with $115 \leq r \leq 146$; this is why the core is not retargeted with the rest.

6.7 The four regions cover the domain

Proposition 6.5 (the four regions cover the domain). *Every integer cell with $r \geq 4$ and $2 \leq t \leq r-2$ lies in at least one of: the finite core (42); the columns $t \in \{2, \dots, 8\}$ of Proposition 6.1; the top region (41); or the joint window (39).*

Proof. For $4 \leq r \leq 503$ use the finite core. Suppose $r \geq 504$. The columns $2 \leq t \leq 8$ are closed by Proposition 6.1, so it remains to consider $t \geq 9$. If $t \geq \sqrt{a_r} + 2$, the cell lies in the top region. Otherwise $t < \sqrt{a_r} + 2$, and hence

$$a_r > (t-2)^2 > 5t, \quad (t-2)^2 - 5t = (t-9)^2 + 9(t-9) + 4 > 0.$$

Thus $r \geq 504$, $t \geq 9 \geq 6$, $a_r \geq 5t$ and $t < \sqrt{a_r} + 2$: all four hypotheses of (39) hold, so the cell is in the joint window. The split uses $\geq$ for the top region and $<$ for its complement, so no equality cell is omitted. The boundary calculation is genuine: at $t = 8$ the same polynomial equals -4 , which is why the seventh fixed column is needed. $\square$

Proof of Theorem 1.1. Let $r \geq 2$ and $2 \leq t \leq r$. If $t = r$ or $t = r - 1$, use Proposition 6.4; the second case can occur only for $r \geq 3$. Otherwise $2 \leq t \leq r - 2$, which forces $r \geq 4$, and Proposition 6.5 places the cell in one of the four interior regions. On the finite core it is the exact certificate of §6.4, used only on its literal integer box. On the columns $t \in \{2, \dots, 8\}$ it is Proposition 6.1. On the top region it is Theorem 6.3. On the joint window it is Theorem 6.2 together with the implication $(36) \Rightarrow \rho(r, t) \geq 1$ of §5.3. $\square$

Lemma 6.6 (orientation of the decrement). *For every $r \geq 2$ and $2 \leq t \leq r$, one has $d_{r,t} > 0$.*

Proof. For $t = r - 1, r$ this is explicit in Proposition 6.4. Suppose $3 \leq t \leq r - 2$. Then $k = t - 2$ satisfies

$$0 < \frac{k\ell}{r} \leq \frac{t-2}{r-1} < 1$$

by Lemma 4.2. Equation (37) therefore gives $\sigma < \mathbf{s}$. Since $0 < \varepsilon < 1$, equations (32) and (33) give $\psi > 0$ and hence $d_{r,t} > 0$.

It remains only to orient the column $t = 2$, because the square-tail ratio itself forgets the sign. The fixed-column tail construction supplies a positive-coefficient lower polynomial for the unsquared decrement on $r \geq 313$. The finite gap $4 \leq r \leq 312$ is rebuilt from (2): after the factorials are cleared, all 309 integer numerators are positive, the first being 15,995,366 at $r = 4$. The two earlier cells $(2, 2), (3, 2)$ are terminal cases already covered above. $\square$

7 The shift-wall application

The square-tail inequality also settles a boundary estimate for the companion shift-wall rows. We state it without that program's internal vocabulary.

7.1 The bridge theorem

Write

$$G = (1 + z)M_{2r-2}, \quad C = M_{2r-1},$$

for the wall rows (7), and consider the inequality

$$(2r)^2 L(G)_q \geq 2 L(C)_q. \quad (44)$$

Theorem 7.1. *Inequality (44) holds for every integer $r \geq 2$ and every integer $0 \leq q \leq r - 2$.*

Proof. For $0 \leq t \leq r$ put

$$O_{r,t} = [z^{r-t}]M_{2r-1}, \quad E_{r,t} = [z^{r-t}]M_{2r},$$

with out-of-range coefficients equal to zero, and normalize

$$P_r(u) = \sum_t \frac{O_{r,t}}{O_{r,0}} u^t, \quad Q_r(u) = \sum_t \frac{E_{r,t}}{E_{r,0}} u^t, \quad x_r = \frac{E_{r-1,0}}{O_{r,0}} > 0.$$

The wall recurrence gives

$$P_r = P_{r-1} + u x_r Q_{r-1}. \quad (45)$$

We first record the only inequality used to iterate (45). If two real-rooted polynomials f, g are in proper position, then every pencil $f + vg$ is real-rooted. Newton's inequality makes $L(f + vg)_t \geq 0$ for every real v . Its discriminant therefore gives

$$B(f, g)_t^2 \leq L(f)_t L(g)_t, \quad B(f, g)_t = f_t g_t - \frac{f_{t-1} g_{t+1} + g_{t-1} f_{t+1}}{2},$$

and polarization yields the Turán triangle

$$\sqrt{L(f + g)_t} \leq \sqrt{L(f)_t} + \sqrt{L(g)_t}. \quad (46)$$

The polynomials P_{r-1} and uQ_{r-1} are positive rescalings and reversals of two consecutive wall rows. To make the interlacing explicit, let $0 < \lambda_1 < \dots < \lambda_{r-1}$ and $0 < \nu_1 < \dots < \nu_{r-1}$ be the positive spectra of J_{2r-2} and J_{2r-1} . Strict Cauchy interlacing and spectral symmetry give

$$0 < \lambda_1 < \nu_1 < \dots < \lambda_{r-1} < \nu_{r-1}.$$

The characteristic-polynomial identity in the proof of Theorem 2.3 makes the zeros of P_{r-1} equal to $-\lambda_j^2$, while those of uQ_{r-1} are the $-\nu_j^2$ together with 0. They strictly interlace, so the two polynomials are in proper position. Thus (45) and (46) give

$$\sqrt{L(P_r)_t} \leq \sqrt{L(P_{r-1})_t} + x_r \sqrt{L(Q_{r-1})_{t-1}}. \quad (47)$$

Set

$$\mathcal{S}_{r,t} = L((1 + u)Q_{r-1})_t, \quad \mathcal{V}_{r,t} = L(Q_{r-1})_{t-1},$$

and

$$\mathcal{A}_{r,t} = r^2 x_r^2 \mathcal{S}_{r,t}, \quad \mathcal{B}_{r,t} = x_r^2 \mathcal{V}_{r,t}, \quad \Delta_{r,t} = \mathcal{A}_{r,t} - \mathcal{A}_{r-1,t}.$$

This is exactly the normalization of Theorem 1.1. Indeed, if $c_r = \Pi_r(0)$ and $\alpha_r = (2r - 2)!!/(2r - 1)!!$, then

$$Q_{r-1} = \Pi_r/c_r, \quad \frac{x_r}{c_r} = \frac{(2r - 1)!}{((2r - 1)!!)^2} = \alpha_r,$$

and

$$\frac{(r - 1)\alpha_{r-1}}{r\alpha_r} = \frac{2r - 1}{2r}, \quad \Delta_{r,t} = r^2 \alpha_r^2 d_{r,t}, \quad \frac{\Delta_{r,t}^2}{2\mathcal{A}_{r,t}\mathcal{B}_{r,t}} = \frac{r^2 d_{r,t}^2}{2\mathcal{S}_{r,t}\mathcal{V}_{r,t}}.$$

Lemma 6.6 supplies $\Delta_{r,t} > 0$. Hence the square-tail inequality, written as $\Delta_{r,t}^2 \geq 2\mathcal{A}_{r,t}\mathcal{B}_{r,t}$, implies

$$r x_r \sqrt{\mathcal{S}_{r,t}} \geq (r - 1) x_{r-1} \sqrt{\mathcal{S}_{r-1,t}} + \frac{x_r}{\sqrt{2}} \sqrt{\mathcal{V}_{r,t}}. \quad (48)$$

To see this, call the first two terms a, b and the last term c . Then $a^2 - b^2 \geq 2ac$ and $a > b$, so $a - b = (a^2 - b^2)/(a + b) \geq c$.

Fix $t \geq 2$. At $r = t$, the polynomial P_{t-1} has degree $t - 1$, so $L(P_{t-1})_t = 0$. Combining (47) and (48) first at $r = t$ and then inductively gives

$$\sqrt{L(P_r)_t} \leq \sqrt{2} r x_r \sqrt{\mathcal{S}_{r,t}} \quad (r \geq t). \quad (49)$$

Finally set $q = r - t$. Reversal of coefficients and the identity $x_r = E_{r-1,0}/O_{r,0}$ give

$$L(P_r)_t = \frac{L(C)_q}{O_{r,0}^2}, \quad x_r^2 \mathcal{S}_{r,t} = \frac{L(G)_q}{O_{r,0}^2}.$$

Squaring (49) is exactly (44). Since $t \geq 2$ is equivalent to $q \leq r - 2$, this proves the stated domain, including $q = 0, 1$. $\square$

Theorem 7.1 closes one leaf of the reduction in the companion spread-at-fixed-product problem [3]; it does not settle that larger problem.

7.2 The lower boundary is exact

On the adjacent stripe $q = r - 1$, that is $t = 1$, an exact streaming sweep over $r = 4, \dots, 3000$ characterizes the stripe completely on that range: the failure set is exactly the contiguous window $r \in [6, 41]$, with minimum ratio

$$0.939129\dots \quad \text{at } r = 13,$$

and the ratio is strictly increasing on $42 \leq r \leq 3000$, so the recovery is monotone. At reverse index 1 the two Turán forms consume only the top four coefficients of each Wall row, which carries the exact sweep far past the full-row range; the streamed values are cross-checked against the full-row computation on $4 \leq r \leq 69$.

For the square-tail inequality itself, the exact deficit at $(r, t) = (2, 1)$ recorded in §1 already prevents a uniform extension to $t = 1$. By contrast, Proposition 6.4 settles $t = r - 1, r$, both with ratio at least 2. Thus $2 \leq t \leq r$ is the natural coefficient range, and its lower endpoint is a boundary of truth.

8 Conclusion

The classical Jacobi/sech and determinantal machinery controls each row of Π_r : real-rootedness, the Poisson–binomial law and the determinant representation. It does not compare adjacent rows.

The cross-row comparison needs three further ingredients: total positivity in the coefficient index, curvature of the lowering profile, and a scalar estimate that keeps its linear and quadratic terms coupled. Together they supply the factor 2 that one-row Newton inequalities miss.

The exact witness $\rho(129, 2) = 1.014576164564870742\dots$ leaves exactly 1.4576% room in the uniform factor: nothing above 2.029152... is admissible. It is the minimum of the certified 500-row core, and by Theorem 1.2 it is the global minimum of ρ on the whole domain, attained there and nowhere else. The coefficient domain itself is sharp at the lower end: $(2, 1)$ fails by the exact deficit $-4351/20736$, while the two terminal cells have ratio at least 2.

Finally, symbolic certificates close the apparent large-row strip. Seven fixed columns leave $t \geq 9$, where the elementary identity $(t - 2)^2 - 5t = (t - 9)^2 + 9(t - 9) + 4$ sends every remaining cell to the top or joint theorem. Exhaustive exact checking is confined to the 500-row core and finite fixed-column prefixes; the terminal cells are symbolic as well.

A The computational record

The proof uses exact computation in two ways. Symbolic certificates — integer or rational identities, coefficient signs and exact Sturm signs — establish the connection-ratio signs of Theorem 5.2, the decrement bound of Theorem 5.4, the Bernstein positivity behind Theorem 5.5, and the coefficient and Sturm certificates of the joint window. Finite exact certificates — integer comparisons, cell by cell — cover the interior core $4 \leq r \leq 503$, the fixed-column prefixes and tails, the terminal replays, and the retargeted chain behind Theorem 1.2. No floating-point value decides anything; reporting decimals are derived after the exact comparisons. Theorem 2.3, Corollary 2.4 and Lemma 5.1 are proved in the text, and their code is transcription replay only.

The companion replay packet documents all of it: every verification script, the exact certificates with their negative controls, the per-result notes, and a single driver, `verify_all.py`,

that rebuilds each certificate from an empty directory and compares it with the released record.

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Large language models were used as mathematical interlocutors: to propose routes, carry out derivations and turn informal intuition into exact statements. Every new mathematical claim was then established by direct proof or an exact certificate; responsibility is mine. The accompanying packet contains the certificates, their verification scripts and the per-result notes.

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